

1. Subdomain Discovery

Tool	Description
Amass ★★★★★	The most comprehensive subdomain enumeration tool. Supports passive and active enumeration, graphing, and asset mapping.
Subfinder ★★★★★	Very fast passive subdomain discovery (ProjectDiscovery).
Assetfinder ★★★★★	Fast and simple passive enumeration.
Findomain ★★★★★	High-speed subdomain discovery with many data sources.
Sublist3r ★★★★★	Older but still useful.
Knockpy ★★★★★	Performs DNS brute-forcing and enumeration.
OneForAll ★★★★★	Powerful Python-based subdomain enumeration tool.

2. DNS Enumeration

Tool	Description
DNSrecon ★★★★★★	Excellent DNS reconnaissance and zone transfer testing.
DNSenum ★★★★★★	Great for DNS enumeration and brute-forcing.
Dig ★★★★★★	Standard Linux DNS lookup utility.
Host ★★★★★★	Simple DNS query tool.
Nslookup ★★★★★	Classic DNS lookup utility.
Fierce ★★★★★	DNS reconnaissance and discovery.

3. DNS Brute Forcing

Tool	Description
ShuffleDNS ★★★★★	Extremely fast DNS brute-forcing (ProjectDiscovery).
MassDNS ★★★★★	Very high-speed DNS resolver.
Gobuster (DNS mode) ★★★★★	DNS brute-forcing using wordlists.
Puredns ★★★★★	High-performance DNS bruteforcer.
DNSenum ★★★	Includes brute-force functionality.

4. Live Host Discovery

Tool	Description
HTTPX ★★★★★	Best tool for checking live HTTP/HTTPS hosts.
httprobe ★★★★★	Lightweight HTTP probing.
cURL ★★★★★	Manual HTTP requests.
Wget ★★★	Retrieve web pages and test connectivity.

5. Port Scanning

Tool	Description
Nmap ★★★★★	Industry standard for port scanning and service detection.
RustScan ★★★★★	Extremely fast port scanner that integrates with Nmap.
Masscan ★★★★★	Internet-scale port scanner.
Naabu ★★★★★	Fast ProjectDiscovery port scanner.
Zmap ★★★★★	Large-scale Internet scanning.

6. Web Technology Detection

Tool	Description
Wappalyzer ★★★★★	Browser extension and CLI for identifying web technologies.
WhatWeb ★★★★★	Detects CMSs, frameworks, and web servers.
BuiltWith ★★★★★	Popular online technology profiler.
HTTPX (-tech-detect) ★★★★★	Detects technologies during probing.

7. Web Application Firewall (WAF) Detection

Tool	Description
WAFW00F ★★★★★	The standard WAF detection tool.
Nmap NSE WAF scripts ★★★★★	WAF detection using Nmap scripts.
WhatWAF ★★★★★	Alternative WAF detection tool.

8. Directory & File Discovery

Tool	Description
Feroxbuster ★★★★★	One of the fastest content discovery tools.
ffuf ★★★★★	Very fast fuzzing and directory enumeration.
Gobuster ★★★★★	Popular directory, DNS, and virtual host brute-forcing tool.
Dirsearch ★★★★★	Python-based directory scanner.
Dirb ★★★★★	Classic directory brute-forcer.

9. Vulnerability Scanning

Tool	Description
Nuclei ★★★★★	Template-based vulnerability scanner.
Nikto ★★★★★	Web server vulnerability scanner.
OpenVAS (Greenbone) ★★★★★	Comprehensive vulnerability assessment platform.
Nessus ★★★★★	Commercial vulnerability scanner used by many organizations.
OWASP ZAP ★★★★★	Web application security scanner and proxy.

10. Web Crawling

Tool	Description
Katana ★★★★★	Modern web crawler by ProjectDiscovery.
Hakrawler ★★★★★	Fast URL discovery tool.
Gau (GetAllURLs) ★★★★★	Retrieves archived URLs from multiple sources.
Waybackurls ★★★★★	Fetches URLs from the Wayback Machine.

11. Screenshot Automation

Tool	Description
EyeWitness ★★★★★	Screenshots websites and identifies technologies.
Aquatone ★★★★★	Website screenshots and reconnaissance.
gowitness ★★★★★	Modern screenshot utility.

12. URL & Parameter Discovery

Tool	Description
ParamSpider ★★★★★	Finds URLs with parameters.
Arjun ★★★★★	Discovers hidden HTTP parameters.
Gau ★★★★★	Retrieves historical URLs.
Waybackurls ★★★★★	Collects archived URLs.

13. Information Gathering (OSINT)

Tool	Description
theHarvester ★★★★★	Collects emails, subdomains, hosts, and employee names.
Recon-ng ★★★★★	Modular OSINT framework.
SpiderFoot ★★★★★	Automated OSINT platform.
Maltego ★★★★★	Visual link analysis and OSINT.

14. Password Attacks

Tool	Description
Hydra ★★★★★	Network login brute-force tool.
John the Ripper ★★★★★	Password cracker.
Hashcat ★★★★★	GPU-accelerated password recovery.
Medusa ★★★★★	Parallel login brute-forcer.

15. Packet Analysis

Tool	Description
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Wireshark ★★★★★	Industry-standard packet analyzer.
tcpdump ★★★★★	Command-line packet capture.
TShark ★★★★★	Command-line version of Wireshark.
Zeek ★★★★★	Network traffic analysis framework.

16. Exploitation

Tool	Description
Metasploit Framework ★★★★★	The most widely used exploitation framework.
Searchsploit ★★★★★	Searches Exploit-DB locally.
Commix ★★★★★	Automated command injection exploitation.
SQLmap ★★★★★	Automated SQL injection testing and exploitation.